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# Virtual Memory and Paging

**CSA1214 Computer Architecture**

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# Introduction

- ▶ Virtual memory is a memory management technique and used by the operating system.
- ▶ It allows programs larger than RAM to run.
- ▶ Uses hard disk as an extension of RAM.
- ▶ Improves multitasking.
- ▶ It helps the computer run many programs at the same time.

## **Example:**

- ▶ Uses disk space when RAM is full. If your computer has **8 GB RAM** but a program needs **10 GB**, virtual memory uses **2 GB from the hard disk**.

# Why Virtual Memory is Needed

## **Problem:**

- ▶ Modern Applications or Large programs need more memory.
- ▶ RAM has limited storage.
- ▶ Running many programs together can fill the RAM.
- ▶ Without enough memory, programs may become slow or stop.

## **Solution:**

- ▶ Virtual memory loads only required pages.
- ▶ Virtual memory stores less-used data on the hard disk.
- ▶ This allows programs to continue running smoothly.

# How Virtual Memory Works

- ▶ The CPU requests data.
- ▶ The Memory Management Unit (MMU) checks the address.
- ▶ If the data is available in RAM, it is used immediately.
- ▶ If the data is not in RAM, it is loaded from the hard disk.
- ▶ The operating system updates the page table.
- ▶ The program continues without restarting.

# Paging

- ▶ Virtual memory is divided into fixed-size blocks called Pages.
- ▶ Memory is divided into Pages.
- ▶ Physical memory (RAM) is divided into Frames.
- ▶ A page table maps pages to frames.
- ▶ Paging reduces external fragmentation.
- ▶ It helps use memory more efficiently.

# Page Fault

- ▶ A page fault occurs when the required page is not present in RAM.
- ▶ The operating system finds the page on the hard disk.
- ▶ The page is copied into RAM.
- ▶ The page table is updated.
- ▶ The CPU continues program execution.

**Note:** Too many page faults make the system slow.

# Difference Between Virtual Memory and Paging

| Virtual Memory                       | Paging                                 |
|--------------------------------------|----------------------------------------|
| It is a memory management technique. | It is a method used in virtual memory. |
| Uses hard disk as extra memory.      | Divides memory into pages and frames.  |
| Helps run large programs.            | Maps pages into RAM frames.            |
| Improves multitasking.               | Reduces external fragmentation.        |
| Depends on paging.                   | Works as part of virtual memory.       |

# Advantages and Applications

## **Advantages:**

- ▶ Allows large applications to run.
- ▶ Improves memory utilization.
- ▶ Supports multitasking.
- ▶ Reduces program crashes due to low memory.
- ▶ Makes better use of available RAM.
- ▶ Improves overall system performance.

## **Applications:**

**Virtual memory and paging are used in:**

- ▶ Windows
- ▶ Linux
- ▶ Android
- ▶ Cloud servers



# Conclusion

- ▶ Disk is slower than RAM.
- ▶ Thrashing reduces performance.
- ▶ Requires MMU and page tables.
- ▶ Virtual memory solves the problem of limited RAM.
- ▶ Paging divides memory into pages and frames for efficient management.
- ▶ These techniques improve multitasking and memory utilization.
- ▶ Every modern operating system uses virtual memory and paging.
- ▶ They are important for fast and efficient computer performance.

THANK YOU